mathematics

MDPI

Article

Location-Routing Optimization for Two-Echelon Cold Chain Logistics of Front Warehouses Based on a Hybrid Ant Colony Algorithm

Xuya Zhang, Yue Wang and Dongqing Zhang *

College of Information Management, Nanjing Agricultural University, Nanjing 210031, China; 31321228@stu.njau.edu.cn (X.Z.); 31321211@stu.njau.edu.cn (Y.W.)
* Correspondence: zhangdq@njau.edu.cn

Abstract: Diverse demands have promoted the rapid development of the cold chain logistics industry. In the paper, a novel approach for calculating the comprehensive carbon emission cost was proposed and the front warehouse mode was analyzed under the background of energy conservation and emission reduction. To solve the two-echelon low-carbon location-routing problem (2E-LCLRP), a mathematical model considering operating cost, total transportation cost, fixed cost, refrigeration cost, cargo damage cost, and comprehensive carbon emission cost was proposed to determine the minimum total cost. A hybrid ant colony optimization (HACO) algorithm based on an elbow rule and an improved ant colony optimization (IACO) algorithm was proposed to solve the 2E-LCLRP. According to the elbow rule, the optimal number of front warehouses was determined and an IACO algorithm was then designed to optimize vehicle routes. An adaptive hybrid selection strategy and an optimized pheromone update mechanism were integrated into the HACO algorithm to accelerate convergence and obtain global optimal solutions. The proposed model and algorithm were verified through the case study of the 2E-LCLRP in Nanjing, China. The HACO algorithm outperformed the original ant colony optimization (ACO) algorithm in terms of convergence rate and solution quality. This study provides significant insights for enhancing heuristic algorithms as well as valuable research methods. Furthermore, the results can help cold chain logistics companies in balancing economic costs and environmental benefits and address cold chain distribution of agricultural products.

Keywords: cold chain logistics; location-routing problem; hybrid ant colony optimization; front warehouse mode; carbon emission

MSC: 90C90

check for updates

Citation: Zhang, X.; Wang, Y.; Zhang, D. Location-Routing Optimization for Two-Echelon Cold Chain Logistics of Front Warehouses Based on a Hybrid Ant Colony Algorithm. *Mathematics* **2024**, *12*, 1851. https://doi.org/10.3390/math12121851

Academic Editors: Shiv Raj Singh, Dharmendra Yadav and Himani Dem

Received: 30 May 2024
Revised: 11 June 2024
Accepted: 12 June 2024
Published: 14 June 2024

Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Due to economic development and the improvement in the quality of life, the demands for the diversity, quality, nutrition, and freshness of foods have largely increased [1]. Huajing Industry Research Institute indicated that the market penetration rate of Chinese online fresh foods had increased from 4.1% in 2017 to 14.6% in 2022, and would further increase. More agricultural products are increasingly transferred to e-commerce platforms, and e-retailers of fresh foods have become important retailers [2]. The competition of fresh products has shifted from low price to high quality and good logistics services, thereby advancing the cold chain logistics industry [3]. The advancement of cold chain logistics has accelerated economic growth, but it has also caused more environmental pollution [4]. Compared with traditional logistics, cold chain logistics is heavily dependent on refrigeration equipment, which increases energy consumption and carbon emissions and thus intensifies the global greenhouse effect and air pollution. Carbon emissions from transportation sectors account for approximately 24.34% of global carbon emissions [5]. Consequently, optimizing the location and distribution strategies of logistics is important

for enhancing logistical efficiency and reducing carbon emissions. Recently, the problems of facility location and route optimization of cold chain logistics have been widely studied [6,7]. The continuous innovation of cold chain logistics has given birth to the novel business mode of front warehouses. This mode employs a two-echelon network structure based on a framework of urban central warehouses, front warehouses, and customers to ensure the timely delivery of goods. The mode increases logistical efficiency, reduces response time, and thereby boosts customer satisfaction [8]. However, compared with other modes, the mode of front warehouses faces the challenge of higher cost, which limits its application scope. Therefore, it is necessary to develop appropriate models according to the actual situation [9]. Especially, joint optimization strategies for location and routing problems should be designed based on low carbon emissions under front warehouses to achieve both economic and environmental benefits. Based on the consideration of operating cost, total fixed cost, refrigeration cost, total transportation cost, cargo damage cost, and comprehensive carbon emission cost, the location-routing model was proposed to minimize the total logistics cost and achieve low-carbon economy. The comprehensive carbon emission cost includes the direct carbon emission cost from fuel consumption and the indirect carbon emission from electric vehicles. The ant colony optimization (ACO) algorithm has been widely used due to its performance in solving complex optimization problems. However, original ACO often cannot yield the best global solution in complex cases. This study proposed a hybrid ant colony optimization (HACO) algorithm, which outperforms ACO in solution quality. HACO incorporates an adaptive hybrid probabilistic decision mechanism with revised pheromone update rules and can avoid early convergence by balancing exploration and exploitation. Despite heuristic limitations and parameter dependency, HACO demonstrated a robust performance and provided excellent approximate solutions with flexibility and scalability in this study.

This paper primarily investigated the front warehouse location-routing optimization problem in a two-echelon transportation network in cold chain logistics based on the consideration of carbon emissions and vehicle types. Related studies focused on location-routing optimization problem models [10], green logistics considering carbon emissions [11], and optimization algorithms [12].

In 1973, Watson-Gandy & Dohrn [13] proposed the joint location-routing problem (LRP) for the first time. LRP has gradually become a research hotspot in the field of logistics optimization. Numerous LRP variants have been proposed, such as dynamic LRP [14], multi-cycle LRP [15], and stochastic LRP [16,17]. Current studies on LRP have focused on the upstream and midstream of the supply chain [18]. In contrast, the last-mile distribution has seldom been reported and has been mainly explored from the perspective of one-echelon LRP [19]. In recent years, multi-echelon logistics networks in the supply chain have been explored. Sun et al. [20] indicated that these networks had a complex dumbbell-shaped structure and were characterized by the cooperative and competitive relationships among echelons. Dai et al. [21] investigated four types of LRP, from one-echelon LRP to four-echelon LRP. Especially, the two-echelon LRP has been the most extensively explored. Mirhedayatian et al. [22] studied the two-echelon distribution system problem and Fazayeli et al. [23] focused on multi-modal transportation logistics and constructed an integer linear planning model to optimize the two-echelon logistics process of supplier, distribution central, and retailer. Darvish et al. [24] explored the flexible two-echelon location-routing problem.

The studies on two-echelon location-routing optimization have focused on the distribution centers as intermediate nodes, but related studies with front warehouses as intermediate nodes have seldom been reported. Along with the rise of e-commerce of fresh agricultural products, the business mode of front warehouses is increasingly integrated into logistics networks due to its advantages of short delivery time and high flexibility [25]. Wu [26] proposed a new front warehouse mode to optimize fresh e-commerce delivery and evaluated its strengths, weaknesses, opportunities, and threats with SWOT and analytic hierarchy process analysis. Zhu & Tian [27] further investigated the value of

high-quality distribution in the front warehouse mode and emphasized the importance of logistics satisfaction and delivery timeliness. Tang et al. [28] developed an integrated optimization model to explore the application of front warehouses in e-commerce. Front warehouses were often located in the downstream of the supply chain. Li & Yang [9] and W. Chen et al. [29], respectively, studied the distribution of large household appliances and the optimization of vehicle routes under the front warehouse mode and constructed corresponding mixed-integer planning models.

In addressing the challenges of climate change and environmental pollution, the sustainable development of the logistics industry is significant. Many scholars investigated the concept and applications of green logistics from the perspective of environmental protection and explored key strategies such as eco-friendly packaging boxes [30], green logistics performance [31], and circular economy [32]. Yao et al. [33] indicated that, along with the concept of sustainable development, the carbon emissions from the logistics industry were increasingly a concern. Cold chain logistics, in particular, generally produces higher carbon emissions than ambient temperature logistics [34]. Therefore, it is significant to develop effective location and routing strategies to reduce carbon emissions. Z. Wang et al. [35] proposed a cold chain logistics location-route model considering fuel consumption, load capacity, and driving distance. Based on the comprehensive modal emission model (CMEM), Dukkanci et al. [36] calculated fuel consumption and carbon costs with vehicle parameters such as friction coefficients, speed, and load. Similarly, with CMEM, C. Zhang et al. [37] calculated fuel consumption and emissions and solved the time-dependent green location-routing problem with time windows (TDGLRP). Furthermore, Li et al. [38] and S. Wang et al. [39] examined transportation-related carbon costs based on the consideration of the policies of carbon tax and quota frameworks.

In the pursuit of low-carbon logistics, in addition to direct carbon emission factors, biofuels and transportation technologies are crucial for emission reduction [40,41]. Additionally, electric vehicles (EVs) are widely considered as they can reduce the dependence on fossil fuels and contribute to the low-carbon circular economy [42,43]. Nilsson & Nykvist [44] analyzed the impact of EVs on low-carbon transportation. T. Wu et al. [45] proposed a planning model for multistage low-carbon EV charging facilities considering carbon price and cruising range to optimize travel routes of EVs. Almouhanna et al. [46] explored the feasibility of replacing fuel vehicles with EVs and solved vehicle routing problems with heuristic algorithms. Furthermore, in order to solve the problems of short range, long charging time, and insufficient charging infrastructure for EVs, Çatay & Sadati [47] introduced an EV routing model and facilitated instant charging of EVs through battery swapping. The above studies provide the theoretical basis and guidance for the application of EVs in logistics and promote the low-carbon and sustainable development of the logistics industry.

Exact algorithms and heuristic algorithms have been employed to solve facility location and routing problems. Exact algorithms can give the optimal solutions to small-scale problems [48], such as the branch-and-price algorithm [49,50], branch-and-bound algorithm [51–53], and Lagrangian relaxation algorithm [54]. Exact algorithms are not applicable to large-scale problems due to their high computational complexity, so heuristic algorithms, especially mixed ones with diverse strategies, are often used in order to overcome the limitations of single algorithms. Biuki et al. [55] used a parallel hybrid of genetic and particle swarm algorithms to solve a multi-objective mixed-integer planning model, and found that the parallel algorithm outperformed serial algorithms.

ACO is a popular heuristic algorithm for complex optimization problems like TSP [56–58] and VRP [59,60], and well known for its global search capability and robustness. Original ACO has been improved by the introduction of new pheromone update rules [61–64] and hybridization with other optimization algorithms [65–67]. Various methods have been explored to prevent algorithms from converging to local optima [68–71]. He et al. [72] developed an adaptive variable neighborhood search ACO and Liang et al. [73] introduced the sub-path support degree to improve ACO performance. Improved ACO performs better in solving many problems. ACO has become a crucial tool for handling

complex optimization issues. Although the improvement strategies have solved certain problems of ACO, it still requires to be further optimized.

In short, front warehouses in two-echelon logistics networks have seldom been explored, particularly the optimization of location and routing. Most studies have focused on direct fuel consumption-related carbon emissions. In this paper, a new method is proposed to calculate indirect carbon emissions of EVs for final-mile delivery and a 2ELC-LRP model considering the front warehouse structure is constructed. Additionally, an improved ant colony algorithm (HACO) based on an elbow rule, an improved probabilistic decision strategy, and a new pheromone update rule is proposed. Finally, a case study demonstrates the effectiveness of the proposed model and HACO algorithm.

2. Mathematical Models

This study focuses on optimizing facility location and vehicles routing for a two-echelon cold chain logistics network in Nanjing to meet customer demands, find the best forward warehouse locations, and plan delivery routes at the minimum total costs and carbon emissions. In this section, an application of 2E-LCLRP in Nanjing is described in detail and a location-routing optimization model is established for this purpose.

2.1. Problem Description

In this paper, a case study of 2E-LCLRP was conducted in Liuhe District, Nanjing, Jiangsu Province (latitudinal range from 32.210053 to 32.527305 and the longitudinal range from 118.577415 to 118.867481). This region was selected due to its typical two-echelon front warehouse network structure and its significant role in the distribution of fresh and perishable goods. In the front warehouse mode of the region, the two-echelon cold chain logistics network for fresh agricultural products is composed of an urban central warehouse, front warehouses, and customers, and each front warehouse serves multiple customers. The same type of refrigerated vehicles are used in the first echelon from a single urban central warehouse to front warehouses, whereas the same type of EVs are used in the second echelon from front warehouses to customers according to the distribution relationship between them. The load weight of vehicles gradually decreases after serving each customer point. When the accumulated customer demand reaches the vehicle's maximum load capacity, the current route is recorded and a new vehicle is allocated to explore a new route until all customer demands are met. EVs return to the original front warehouse after delivery. The problem of 2E-LCLRP is illustrated in Figure 1.

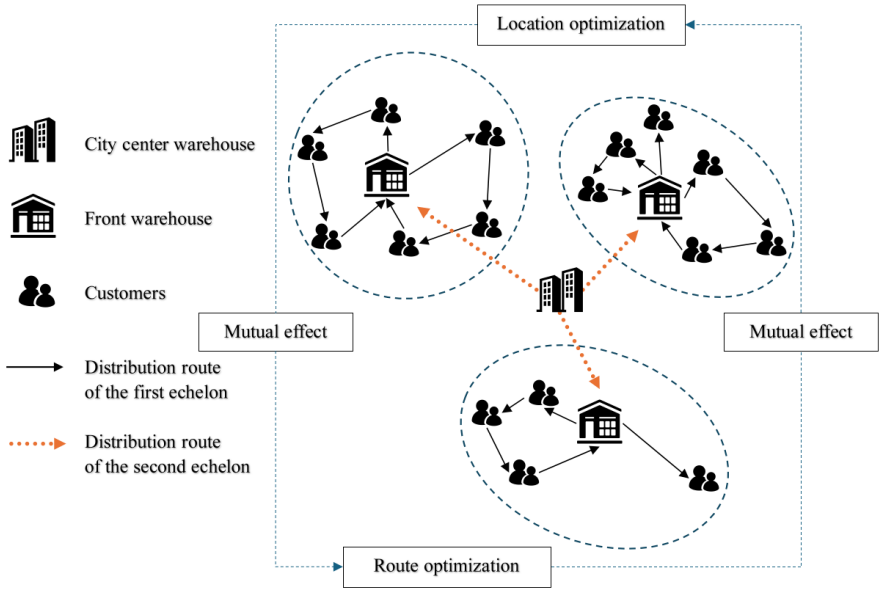


Figure 1. Illustration of a 2E-LCLRP model in Liuhe District, Nanjing, Jiangsu Province, China.

2.2. Problem Assumptions and Notations

In this section, relevant notations are defined in Table 1 and reasonable assumptions of the 2E-LCLRP model are provided as follows.

Assumption 1. *The locations of the urban central warehouse, front warehouses, and customers are known and the demand of customers is known [74].*

Assumption 2. *The types of delivery demands of all customers are the same [75].*

Assumption 3. *Each refrigerated vehicle starts its route from the same single urban central warehouse and each electric vehicle departs from one front warehouse and finally returns to the same front warehouse [25].*

Assumption 4. *Each customer is served once by only one front warehouse and one vehicle [9].*

Assumption 5. *Vehicles travel with a known average speed without considering road and traffic conditions [19].*

Table 1. Notations and descriptions in the 2E-LCLRP model.

Notations	Descriptions
N	A set of logistics facilities including the urban central warehouse and front warehouses, $N = \{0, 1, 2, 3, \ldots, n\}$ and 0 denotes the urban central warehouse
M	A set of customers, $M = \{0, 1, 2, 3, \ldots, m\}$
S	A set of refrigerated vehicles for transportation routes in the first echelon, $S = \{0, 1, 2, 3, \ldots, s\}$
V	A set of electric vehicles for distribution routes in the second echelon, $V = \{0, 1, 2, 3, \ldots, v\}$
Z_n	If front warehouse is selected to be opened, $Z_n = 1$; otherwise, $Z_n = 0$, $n \in N$
Z_{0ns}	If refrigerated vehicle s travels from urban central warehouse 0 to front warehouse n in the first echelon, $Z_{0ns} = 1$; otherwise, $Z_{0ns} = 0$, $n \in N$, $s \in S$
Z_{nmv}	If electric vehicle v travels from front warehouse n to customer m in the second echelon, $Z_{nmv} = 1$; otherwise, $Z_{nmv} = 0$, $n \in N$, $m \in M$, $v \in V$
Z_{nm}	If front warehouse n is selected for serving customer m, $Z_{nm} = 1$; otherwise, $Z_{nm} = 0$, $n \in N$
A_n	Operating cost of front warehouse n, $n \in N$
B_s^1	Fixed cost for transportation by refrigerated vehicle s, $s \in S$
B_v^2	Fixed cost for distribution by electric vehicle v, $v \in V$
C_1	Distribution cost per unit of distance and weight for refrigerated vehicle s in the first echelon, $s \in S$
C_2	Distribution cost per unit of distance and weight for electric vehicle v in the second echelon, $v \in V$
l_{0n}	Distance in the first echelon from urban central warehouse to front warehouse n, $n \in N$
l_{nm}	Distance in the second echelon from front warehouse n to customer m, $n \in N$, $m \in M$
p_0	Unit price of fresh agricultural products
λ	Freshness attenuation coefficient
W_{0n}^s	Load weight of refrigerated vehicle s from the urban central warehouse 0 to front warehouse n, $n \in N$, $s \in S$
W_{nm}^v	Load weight of electric vehicle v from front warehouse n to customer m, $n \in N$, $m \in M$, $v \in V$
t_{0n}	Travel time from the urban central warehouse to front warehouse n, $n \in N$
t_{nm}	Travel time from front warehouse n to customer m, $n \in N$, $m \in M$
E_1	Direct carbon emission coefficient
E_2	Indirect carbon emission coefficient
p_c	Unit carbon tax price
f	Refrigerant consumption cost per unit of time
$E(W_{0n})$	Relationship between fuel consumption and load capacity per unit of distance
ε_m	Fuel consumption per unit of distance for refrigerated vehicle s while fully loaded, $s \in S$
ε_0	Fuel consumption per unit of distance for refrigerated vehicle s while unloaded, $s \in S$
e_0	Electrical energy consumption per unit of distance for electric vehicle v, $v \in V$
Q_s^1	Maximum capacity of refrigerated vehicles
Q_v^2	Maximum capacity of electric vehicles
q_n	Capacity of front warehouse n, $n \in N$
q_m	Demand of customer m, $m \in M$

2.3. The Model of Two-Echelon Low-Carbon Location-Routing Problem

This paper investigated the two-echelon low-carbon location-routing problem (2E-LCLRP) model of cold chain logistics of front warehouses, referred to as the 2E-LCLRP model. In the model, the total cost is composed of operating cost, transportation cost, fixed cost, refrigeration cost, cargo damage cost, and comprehensive carbon emission cost. The objective function of the model is the minimal total cost.

2.3.1. Operating Cost

The operational cost of front warehouses is composed of the rental fee, equipment expense, personnel management cost, and other related expenses. The operational cost varies with the leased area of front warehouses. Operating cost is expressed as:

$$C_1 = \sum_{n}^{N} A_n Z_n. \tag{1}$$

2.3.2. Total Fixed Cost

Total fixed cost in distribution operation is the expenses, which are not related to driving distance or time, and involves key items such as labor, equipment depreciation, and vehicle maintenance. In this paper, the fixed cost for transportation in the first echelon is different from that in the second echelon. Hence, the total fixed cost in cold chain logistics is computed as:

$$C_2 = \sum_{s}^{S} \sum_{n}^{N} B_s^1 Z_{0ns} + \sum_{v}^{V} \sum_{n}^{N} \sum_{m}^{M} B_v^2 Z_{nmv}. \tag{2}$$

2.3.3. Total Transportation Cost

Transportation cost is related to the expense of transporting fresh products and generally positively correlated with driving distance. In a two-echelon logistics network of front warehouses, transportation in the second echelon typically involves the itinerant distribution mode. Total transportation cost is computed as:

$$C_3 = \sum_{s}^{s} \sum_{n}^{N} C_1 I_{0n} Z_{0ns} + \sum_{v}^{V} \sum_{n}^{N} \sum_{m}^{M} C_2 I_{nm} Z_{nmv}. \tag{3}$$

2.3.4. Refrigeration Cost

Refrigeration cost refers to the refrigerant consumption cost for maintaining vehicle temperature in the cold chain logistics and is positively correlated with travel time. The cost is only related to refrigeration transportation in the first echelon as EVs used in the second echelon lack the refrigeration function. Refrigeration cost is computed as:

$$C_4 = \sum_{s}^{s} \sum_{n}^{N} f t_{0n} Z_{0ns}. \tag{4}$$

2.3.5. Cargo Damage Cost

Cargo damage cost represents the expense caused by the deterioration of fresh goods. The cost is considered in the first echelon and is proportional to time. Due to the short distance away from the customers in the second echelon, the corresponding cargo damage cost is so small that it can be neglected in the analysis. Cargo damage cost can be expressed as:

$$C_5 = \sum_{s}^{s} \sum_{n}^{N} p_0 \lambda W_{0n}^s t_{0n} Z_{0ns}. \tag{5}$$

2.3.6. Comprehensive Carbon Emission Cost

Against the background of pursuing carbon emission peak and carbon neutrality, direct carbon emissions and indirect emissions are considered in the study. Direct carbon emissions stem from fossil fuel combustion such as coal and oil when refrigerated vehicles travel from an urban central warehouse to a front warehouse. The calculation involves various factors such as travel distance, load capacity, and direct carbon emission coefficient. The relationship between fuel consumption per unit distance and load capacity of vehicles is expressed as:

$$E(W_{0n}) = \frac{\varepsilon_m - \varepsilon_0}{Q_s^1} W_{0n}^s + \varepsilon_0. \tag{6}$$

In the last-mile delivery, EVs are widely utilized to realize the benefit of zero tailpipe emission. EVs generate power from electricity. Since electricity production in China is predominantly generated from fossil fuels, carbon emissions in this process are not negligible. In the calculation of carbon emissions of electric tricycles, carbon cost should be indirectly estimated with the total electrical energy consumption in vehicle operation. An innovative indirect carbon emission calculation formula considering the indirect carbon emission coefficient, electricity consumption per unit distance, and travel distance is introduced below. Based on both direct and indirect carbon emission costs, the comprehensive carbon emission cost is expressed as:

$$C_6 = \sum_{s}^{s} \sum_{n}^{N} Z_{0ns} p_c E_1 E(W_{0n}) l_{0n} + \sum_{v}^{V} \sum_{n}^{N} \sum_{m}^{M} Z_{nmv} p_c E_2 e_0 l_{nm}. \tag{7}$$

Therefore, the formal model for 2E-LCLRP can be defined as:

$$\min_{\mathrm{U}} \mathrm{C} = C_1 + C_2 + C_3 + C_4 + C_5 + C_6, \tag{8}$$

s.t.

$$\sum_{n \in N} Z_n \geq 1, \tag{9}$$

$$q_n Z_n \geq \sum_{m \in M} q_m Z_{nm}, \forall n \in N, \tag{10}$$

$$W_{ij}^v = \sum_{i=1}^{m} \left(W_{nm}^v - \sum_{j=1}^{i=1} q_j \right), \tag{11}$$

$$\sum_{n \in N} \sum_{m \in M} q_m Z_{nmv} \leq Q_v^2, \forall v \in V, \tag{12}$$

$$\sum_{v \in V} Z_{nmv} = 1, \forall m \in M, \tag{13}$$

$$\sum_{m \in M} Z_{nmv} = \sum_{m \in M} Z_{mnv}, \forall v \in V. \tag{14}$$

The objective function (Equation (8)) is to minimize the total costs of both echelons, where U stand for the parameters Z_n, Z_{0ns}, Z_{nmv}, and Z_{nm}. The constraint (Equation (9)) limits the open number of front warehouses. The constraint (Equation (10)) limits the total demand of served customers to the front warehouse's capacity. The constraint (Equation (11)) describes the gradual decrease of a vehicle's load capacity during cargo delivery. The constraint (Equation (12)) guarantees that the total demand of each route should be within the vehicle's load limit. The constraint (Equation (13)) ensures that each customer is served by only one vehicle. The constraint (Equation (14)) indicates that after electric vehicle v leaves from front warehouse n it will return to front warehouse n.

3. Design of HACO Algorithm

Effective problem-solving methods are crucial for improving the overall efficiency and reducing the cost of NP-hard problems. In the study, the HACO algorithm, in which the elbow rule was integrated with IACO, was designed to solve the 2E-LCLRP. In the designed HACO algorithm, the clustering analysis of customer points is firstly performed according to the elbow rule to determine the optimal number of front warehouses. Then, customers are allocated according to distance. Finally, with IACO, the optimal route scheme and objective function are solved.

3.1. Elbow Rule

The key to the elbow rule lies in determining a reasonable number of clusters based on the trend of the sum of squared errors (SSE), where SSE represents the sum of the squared errors of the distances of each point away from its respective cluster central [76]:

$$SSE = \sum_{i=1}^{k} \sum_{p \in C_i} |p - m_i|^2, \tag{15}$$

where k represents the cluster count; C_i represents cluster i; p represents a sample point within a cluster; and m_i represents the mean of all data points within a cluster. With the gradual increase in clusters, the distribution characteristics of the data can be obtained with higher precision, thus leading to a decrease in SSE. When the number of clusters increases to a critical value, the decrease in SSE becomes less pronounced, so that the critical point is manifested as the "elbow" and indicates the optimal number of clusters [77].

3.2. Improved Ant Colony Optimization Algorithm

The IACO algorithm is obtained based on the original ACO framework by adopting the adaptive probabilistic decision strategy and the pheromone update rule to solve the limitations of the original ACO algorithm, including convergence efficiency, local optimal solution, and stagnation.

3.2.1. Original Ant Colony Optimization Algorithm

ACO is a heuristic search algorithm inspired by the foraging behavior of ants in nature [78]. Ants communicate by leaving pheromones along the way, which accumulate and are iteratively updated, so that the ant colony can identify the shortest path from the nest to a food source [79]. The ACO algorithm has been widely applied in solving combinational optimization problems. However, it has a low iteration efficiency and a poor local search ability, which potentially lead to algorithmic stagnation [80].

To refine the ACO algorithm, H. Wu et al. [81] and Chen et al. [82] optimized the probabilistic decision mechanism and pheromone update rules to balance the exploration and exploitation of the algorithm. This optimized approach mentioned only considered the immediate change of the best solution in a single iteration, thus affecting the stability and quality of the solution. In this paper, a dynamic adjustment mechanism is introduced to enhance the adaptive selection ability between random selection and deterministic selection, and the pheromone update rule is adjusted to take advantages of the optimal and sub-optimal paths.

3.2.2. Dynamic Adjustment Mechanism

In this paper, the optimal total cost at each iteration is used as a sequential variable dependent on the number of selected generations. According to Equation (16), in a fixed-length sliding window, the average of the optimal cost is computed from the most recent iterations and compared with the optimal total cost computed from the previous iteration. Based on this comparison result, the dynamic adjustment of r_0 is determined. If the average is less than the optimal cost of the previous iteration, it is suggested that a better solution has been found. In other words, it is proper to increase r_0 to raise the proportion of

deterministic selection. Otherwise, it is proper to decrease the value of r_0 to increase the proportion of random selection. The adjusted value of r_0 is between 0 and 1 to ensure stability. The enhanced dynamic adjustment method is expressed as:

$$C_{avg} = \begin{cases} \frac{1}{w}\sum_{i=t-w}^{t-1} C_{best,i}, & t > w \\ \frac{1}{t}\sum_{i=1}^{t} C_{best,i}, & t \le w \end{cases}, \tag{16}$$

$$r_0 = \begin{cases} r_0(t-1)\left(1 + \frac{C_{avg} - C_{best,t-1}}{C_{best,t>w}}\right), & C_{avg} < C_{best,t-1} \\ r_0(t-1)\left(1 - \frac{C_{avg} - C_{best,t-1}}{C_{best,t-1}}\right), & C_{avg} \ge C_{best,t-1} \end{cases}. \tag{17}$$

3.2.3. Deterministic Selection

When the ant colony selects the next node, the maximum state transition probability among accessible nodes is determined with the pheromone concentration and heuristic function, as expressed in Equation (19). A random parameter in the interval of $[0, 1]$ is generated and then compared with a selection threshold, r_0. If $r \le r_0$, the ant selects the node with the maximum state transition probability as the next node to be visited. This selection strategy is a deterministic strategy as the ant always selects the node with the highest probability:

$$\eta_{ij} = \frac{1}{d_{ij}}, \tag{18}$$

$$j = \text{argmax}_{j \in \text{Tabu}_k}\left(\left[\tau_{ij}(t)\right]^{\alpha}\left[\eta_{ij}(t)\right]^{\beta}\right), \tag{19}$$

where d_{ij} indicates the distance between node i and node j; η_{ij}^{β} represents the heuristic function and is typically inversely proportional to the distance between nodes, as described in Equation (18); τ_{ij}^{α} represents the pheromone concentration on the path between nodes; $Tabu_k$ indicates the set of nodes to be visited; and α and β are the parameters describing the important weights of pheromones and heuristic function factor in the travel route, respectively.

3.2.4. Random Selection

When $r > r_0$, with a roulette wheel selection method, the ant, k, randomly chooses one of the accessible nodes as the next node to be visited, as expressed in Equation (20). This method is a biased random selection method based on the pheromone concentration and the reciprocal of the path length [83]. This biased selection mechanism is still widely utilized in ant colony algorithms, as it leads ants to prefer the paths that have historically performed better [71]. Simultaneously, paths with lower probabilities retain a certain likelihood of being selected, thus maintaining diversity in the search process. Experiments conducted by Chen et al. [82] experimentally validated the efficacy of this selection mechanism. The results indicate that the biased random selection can prevent the algorithm from premature convergence and enhance the quality of the final solution.

$$p_{ij}^{k}(t) = \begin{cases} \frac{\left[\tau_{ij}(t)\right]^{\alpha}\left[\eta_{ij}(t)\right]^{\beta}}{\sum_{j \in Tabu_k}\left[\tau_{ij}(t)\right]^{\alpha}\left[\eta_{ijj}(t)\right]^{\beta}}, j \in Tabu_k \\ 0, else \end{cases} \tag{20}$$

3.2.5. Pheromone Update Rule

In original ACO, ants continuously explore the optimal path in the iterative process, and each visited path undergoes a process of pheromone evaporation and deposition. However, the global pheromone update strategy also leads to resource waste on non-optimal paths and increases execution time. To overcome this limitation, a local pheromone update strategy [84] was proposed to implement pheromone evaporation and enhancement on the optimal path and only enhancement on other paths. This strategy boosted its positive

feedback effect, but it weakened its exploration capability. Moreover, the ant-circle model outperformed the ant-quantity and ant-density models in terms of global search efficiency.

Consequently, the pheromone update mechanism is refined by extending it to sub-optimal paths. In addition, the pheromone increment on the optimal path is larger than that on the sub-optimal path, thus maintaining the preference for the optimal solution:

$$\tau_{ij}(t+1) = \begin{cases} (1-\rho)\tau_{ij}(t) + \Delta\tau_{ij}, (i,j) \in bestR \\ (1-\rho)\tau_{ij}(t) + \Delta\tau_{ij}, (i,j) \in bestR' \\ \tau_{ij}(t)(1-\rho), else \end{cases}, \quad (21)$$

$$\Delta\tau_{ij} = \frac{Q}{L_{best}}, \quad (22)$$

$$\Delta\tau'_{ij} = \frac{Q}{L'_{best}}, \quad (23)$$

where τ_{ij} is the pheromone concentration between node i and node j; ρ is pheromone evaporation coefficient; Q is the pheromone constant; $bestR$ and $bestR'$, respectively, represent the paths taken by the optimal ant and the sub-optimal ant; L_best and L_best', respectively, represent the lengths of the optimal and sub-optimal paths; and $\Delta\tau_{ij}$ and $\Delta\tau'_{ij}$ are the pheromone increments of optimal and sub-optimal paths, respectively.

3.3. Steps of IACO Algorithm

According to the framework of original ACO integrated with the adaptive probabilistic decision strategy and pheromone update rule, IACO is used to optimize the whole vehicle routes in global space [85]. The steps of IACO algorithm are introduced below:

Step 1: parameter initialization. The number of the current iteration is set as $iter$ =1, and $Tabu_k = \emptyset$. Then, the parameter initialization of m, α, β, Q, ρ, r_0, and $iter_{max}$ is implemented.

Step 2: solution space construction. An empty $Tabu_k$ is generated and all ants are randomly placed on candidate front warehouse nodes. Then, customer nodes are assigned to each front warehouse based on distance and capacity.

Step 3: node selection. The next node is determined based on the adaptive probabilistic decision strategy while ensuring that the cumulative customer demands are lower than vehicle capacity. Traversal routes are recorded and the next vehicle is assigned to explore the new route when the cumulative customer demands equal the capacity of the vehicle.

Step 4: $Tabu_k$ is continuously updated and the process is repeated until all customer nodes are visited and vehicles return to front warehouses.

Step 5: recording the optimal solution. The current optimal total cost is compared with the historical optimal solution $C(t-1)$. If $C(t)$ is less than $C(t-1)$, output $C(t)$; otherwise, output $C(t-1)$.

Step 6: updating pheromone. After all nodes are visited, the total path length of each ant is calculated and pheromone concentrations for the optimal and sub-optimal paths are updated with Equation (21) to reflect search performance.

Step 7: termination conditions. If the iteration limit $iter_{max}$ is not reached, the current iteration number is increased by one and $Tabu_k$ is emptied. Then, return to Step 2 to continue the iteration. Otherwise, the calculation is terminated and the optimal solution is output.

The above calculation process is illustrated in Figure 2.

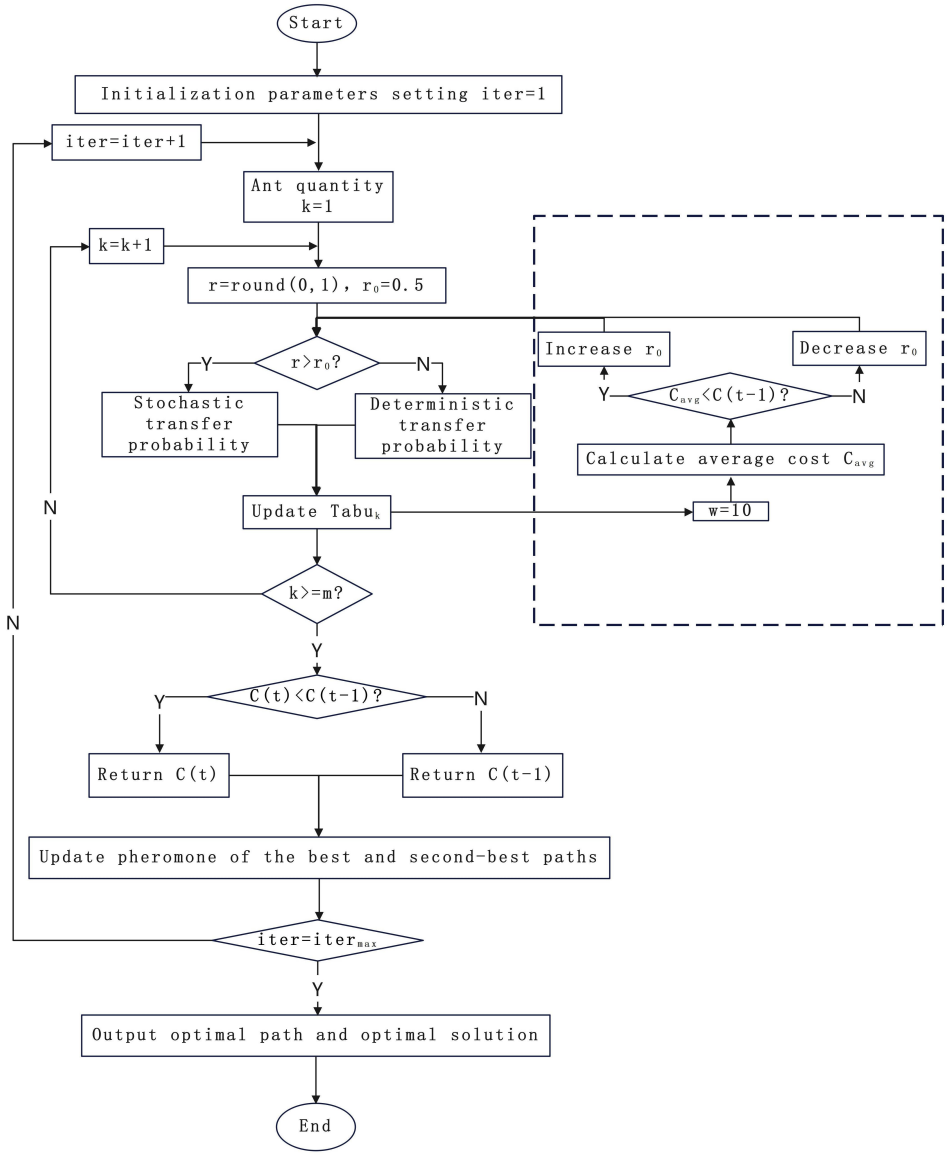


Figure 2. Flow chart of IACO algorithm.

4. Case Study

This section presents a case study that applies HACO to a realistic two-echelon cold chain logistics network. The study aims to address location-routing optimization problems in the context of energy conservation and emission reduction. The section starts with a comprehensive description of the parameter settings and then presents results of the case study. This case study has validated the efficacy of HACO in reducing overall logistics costs and carbon emissions, demonstrating superior performance in terms of solution quality and computational efficiency compared to ACO.

4.1. Parameter Setting

According to the distribution of customers in Liuhe District, a large residential community, to evaluate delivery efficiency comprehensively, the orders from the same community and time period were selected as the research object. Over 100 POI location data were filtered with the API tool of the Gaode Map software version 4.1.3. The location data of a urban central warehouse, six candidate front warehouses (with codes 36 to 41), and 35 customer points were obtained. Figure 3 displays the spatial distribution of the obtained data. The daily demand for fresh agricultural products was estimated based on the permanent

population in residential areas (Table 2). Operational costs and maximum capacities of front warehouses of different sizes were calculated with warehouse rental fee, heights, and the unit weight of fresh products. The relevant data are provided in Table 3.

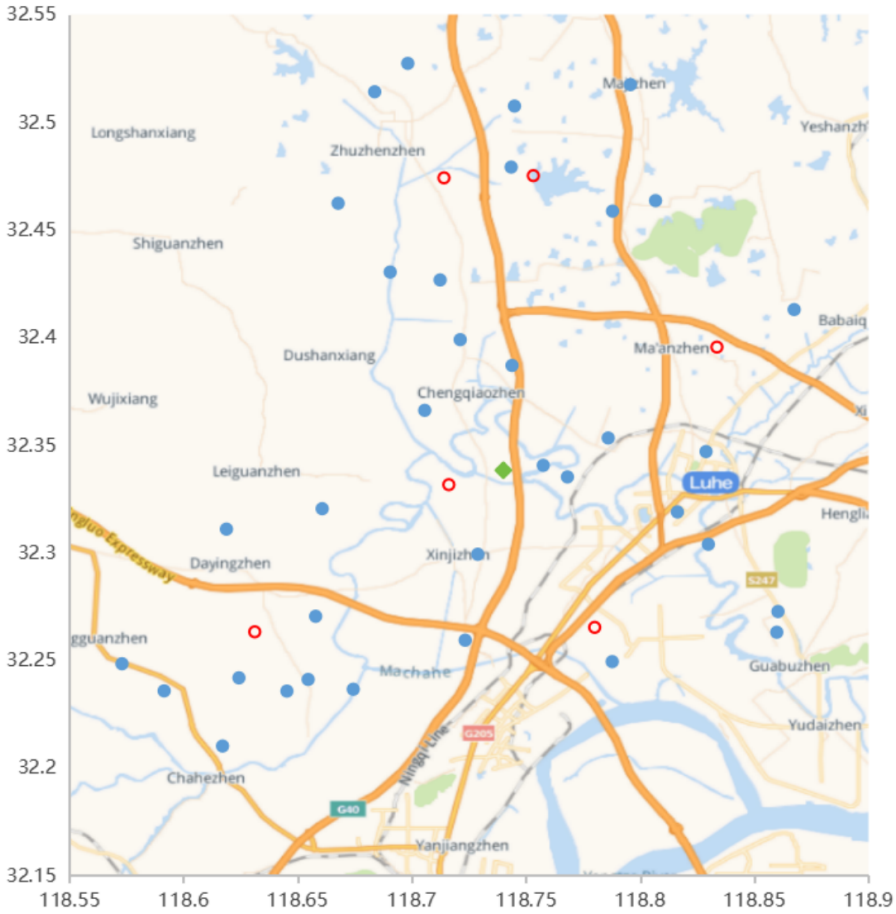


Figure 3. Location situation of the nodes in Liuhe District, Nanjing, Jiangsu Province.

Table 2. Information of customers.

No	Longitudes	Latitudes	Demands/kg
Customer 1	118.674395	32.236451	88
Customer 2	118.816348	32.318846	118
Customer 3	118.860394	32.272571	64
Customer 4	118.822899	32.303817	104
Customer 5	118.743556	32.479112	98
...	...	...	...
Customer 35	118.721234	32.3989120	65

Table 3. Information of an urban central warehouse (UCW) and front warehouse (FW).

No	Longitudes	Latitudes	Operating Costs	Capacity/kg
UCW	118.740120	32.338211	–	–
FW 36	118.753252	32.475142	1083	230,000
FW 37	118.714124	32.474121	1092	322,000
FW 38	118.833643	32.395467	1099	414,000
FW 39	118.783981	32.266512	1073	202,000
FW 40	118.631251	32.263141	1082.5	230,000
FW 41	118.716244	32.331521	1089.1	3,105,000

.

In order to convert latitude and longitude coordinates into quantifiable actual distances, with the Haversine formula [86,87], the shortest distance between two points on the spherical surface is calculated as follows:

$$a = \sin^2\left(\frac{\Delta\, lat}{2}\right) + \cos(\, lat1)\cos(lat2)\sin^2\left(\frac{\Delta\, lon}{2}\right), \tag{24}$$

$$c = 2\,\mathrm{atan}\,2(\sqrt{a}, \sqrt{1-a}), \tag{25}$$

$$d = \; Rc, \tag{26}$$

where $\Delta\, lat$ is the latitude difference; $\Delta\, lon$ is the longitude difference; $lat1$ and $lat2$ represent the latitudinal positions of two points; R represents the radius of the Earth and is about 6371 km on average; and d is the distance between two points.

In this paper, refrigerated vehicles of Foton's European Air R series were selected as the cold chain distribution vehicle in the first echelon, whereas EVs with a battery specification of 60 V 4AH were used in the second echelon. Based on relevant vehicle parameters and the statistical data released by the Industrial Research Institute of China, the parameters are shown in Tables 4 and 5.

Table 4. Information of refrigerated vehicles and electric vehicles.

Parameters	Values of Refrigerated Vehicles	Values of Electric Vehicles
Weight/t	7.875	200
Fixed cost/CNY	300	120
Unit transportation cost/(CNY · km^{-1})	4.5	1.5
Rated load capacity/t	9.995	250
Average traveling speed/(kg · h^{-1})	30	15

Table 5. Information of other parameters.

Parameters	Values
p_0	32/(CNY · kg^{-1})
λ	0.02
f	12/(CNY · h^{-1})
p_c	54.22/(CNY · t^{-1})
E_1	2.63/(kgCO_2e · L^{-1})
E_2	0.95/(kgCO_2e · L^{-1})
ε_m	0.32/(L · km^{-1})
ε_0	0.11/(L · km^{-1})
e_0	0.07/(kWh · km^{-1})

To compare the performances of the original ACO algorithm and HACO algorithm under the optimal conditions, parameter optimization experiments were carried out with the two algorithms. By adjusting α, β, Q, and ρ, the algorithm can balance the exploration and exploitation capabilities and improve the solution quality. Therefore, the algorithm can avoid local optima to converge towards global optima. The optimal parameter configurations for both algorithms are shown in Table 6.

Table 6. Optimal parameter configurations of ACO and HACO algorithms.

Parameters	Values of ACO	Values of HACO
m	50	50
α	2	5
β	2	1
Q	300	700
ρ	0.3	0.3
$iter_{max}$	200	200

4.2. Results

In this paper, according to the elbow rule, the customer point cluster analysis was performed to determine the number of front warehouse locations. SSE decreased gradually when the number of cluster centers increased to five or six (Figure 4), indicating that further increasing cluster centers could not significantly improve the cluster quality. The number of front warehouses was positively correlated with inherent costs such as rent, equipment acquisition and maintenance cost, and management cost, and so the number of cluster centers was set to be five for the purpose of cost reduction and efficiency improvement.

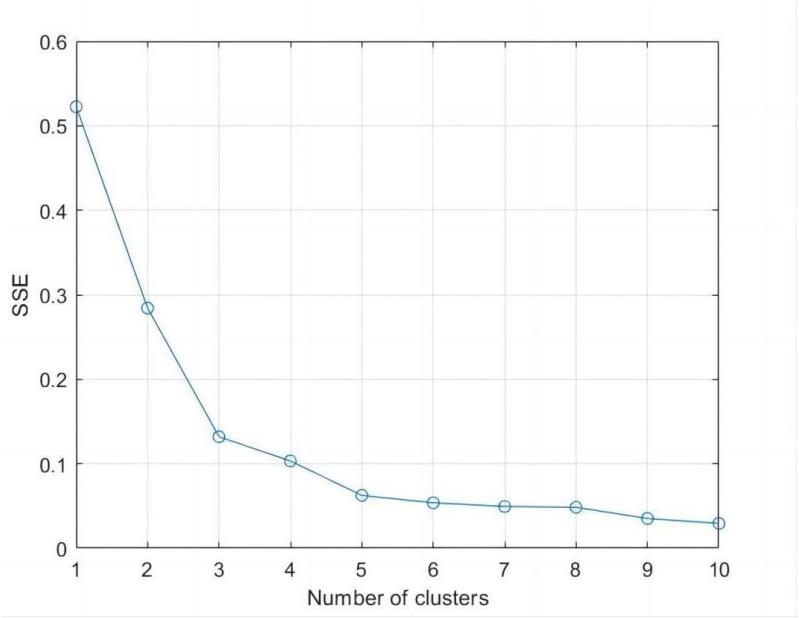


Figure 4. Numbers of the cluster centers obtained with the elbow rule.

To validate the superiority of the HACO algorithm, this study used ACO and HACO algorithms to investigate the case study and the number of iterations was set at 200. The convergence performances of both algorithms on carbon emission cost and total cost are illustrated in Figures 5 and 6. The results indicated that HACO consistently outperformed ACO in the initial stages, indicating its tendency towards the global optimal solution. During the iterative process, HACO demonstrated a significantly faster convergence rate compared with ACO, as indicated in Figure 6. In the mid-stage of algorithm operation, ACO experienced stagnation for over 100 iterations. Conversely, the HACO algorithm effectively combined randomness with determinism and enhanced the algorithm's search capabilities through its adaptive hybrid selection strategy, thus avoiding the local optimal solutions and stagnation observed in ACO. The HACO algorithm converged to the optimal solution of the objective function within 20 iterations, which could be attributed to its refined pheromone update mechanism, which promotes a deeper exploration of the global optimal solution by the algorithm, thereby enhancing the overall convergence efficiency.

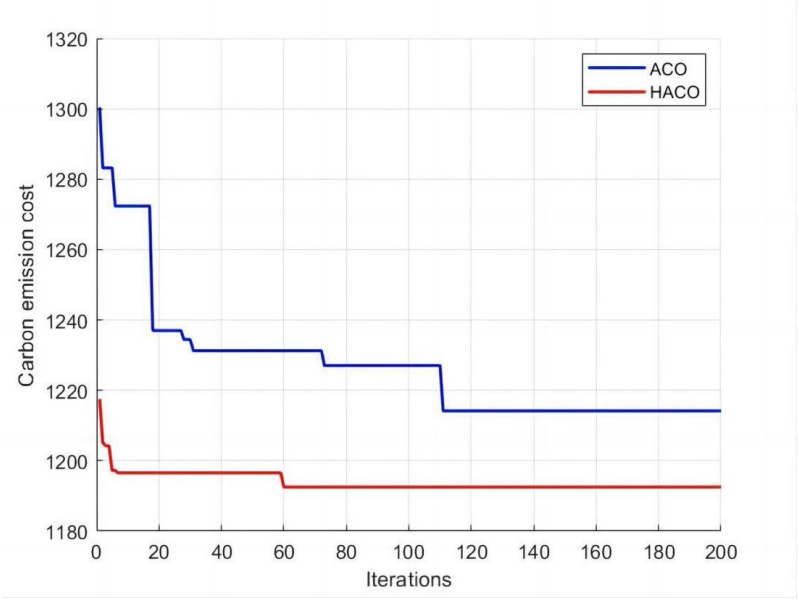


Figure 5. Convergence curves of ACO and HACO on comprehensive carbon emission costs.

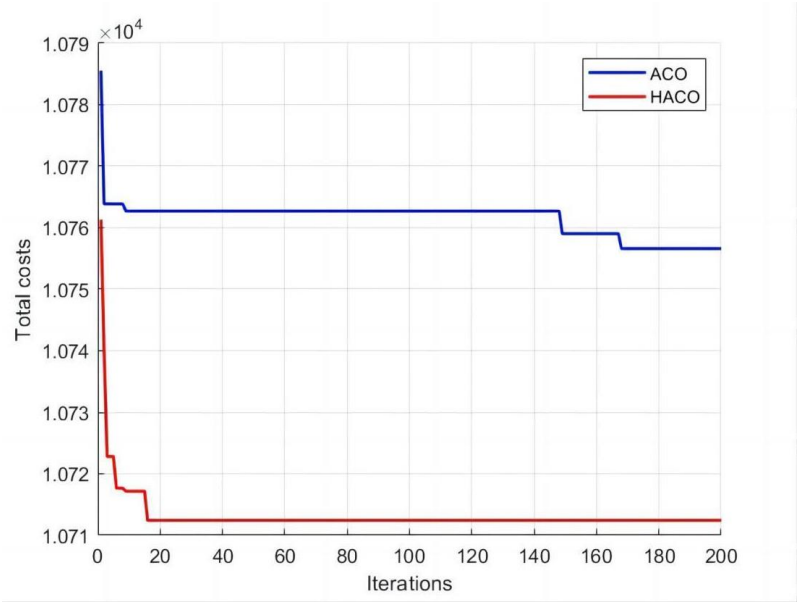


Figure 6. Convergence curves of ACO and HACO on total costs.

Figures 7 and 8, respectively, present the optimal location-routing schemes identified with the ACO and HACO algorithms. Table 7 provides the details of these optimal schemes. The results indicated that the HACO algorithm could identify higher-quality path planning schemes and achieve an optimal solution to the objective function while maintaining consistent site selection strategies. This discrepancy indicated that the path planning scheme maintained its independence to a certain degree although it was affected by location decision. Different execution strategies led to the variations in the exploration of the solution space. The ACO algorithm relied solely on the roulette wheel selection method when the next node was selected, thus limiting the node selection scope and a local optimal solution. In contrast, the HACO algorithm employed an adaptive probabilistic decision strategy that achieved the better balance between exploration and exploitation. This strategy allowed the algorithm to adaptively adjust its search strategy based on the characteristics of the solutions at different iteration stages, potentially revealing path planning schemes that were not explored with traditional ant colony algorithms. Furthermore, the HACO algorithm's local pheromone enhancement mechanism enabled it to more accurately depict

the relatively high-quality solution space, thereby facilitating the discovery of superior path planning schemes.

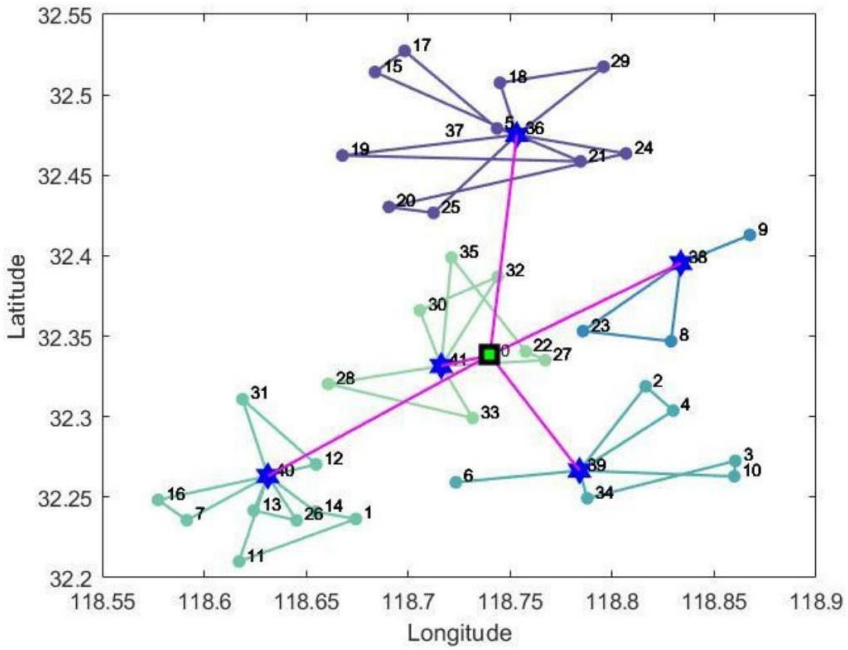


Figure 7. Illustration of the solutions obtained by ACO, depicting three types of nodes by three icons. Homogeneous colored lines depict vehicle routes originating from the same UCW, heterogeneously colored lines depict vehicle routes departing from different FWs.

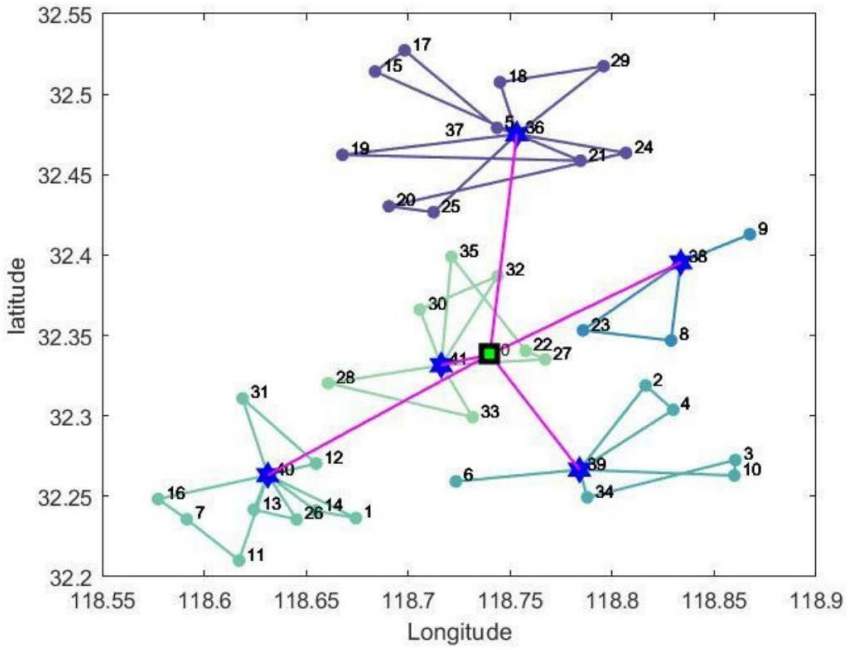


Figure 8. Illustration of the solutions obtained by HACO, depicting three types of nodes by three icons. Homogeneous colored lines depict vehicle routes originating from the same UCW, heterogeneously colored lines depict vehicle routes departing from different FWs.

Table 7. Comparison between the vehicle route schemes obtained with ACO and HACO algorithms.

Types	Locations	Vehicle Route Schemes
ACO	36	36-5-17-15-36, 36-18-29-36, 36-25-20-24-36, and 36-21-19-36
	38	38-23-8-38 and 38-9-38
	39	39-4-2-39, 39-6-39, and 39-10-3-34-39
	40	40-16-7-40, 40-14-1-11-40, 40-13-26-40, and 40-31-12-40
	41	41-27-22-35-41, 41-32-30-41, and 41-33-28-41
HACO	36	36-5-17-15-36, 36-18-29-36, 36-25-20-24-36, and 36-21-19-36
	38	38-23-8-38 and 38-9-38
	39	39-4-2-39, 39-6-39, and 39-10-3-34-39
	40	40-16-7-11-40, 40-14-1-40, 40-13-26-40, and 40-31-12-40
	41	41-27-22-35-41, 41-32-30-41, and 41-33-28-41

Table 8 shows the cost metrics obtained with ACO and HACO algorithms. Compared with original ACO, the HACO algorithm demonstrated the better performance in terms of solution efficiency and quality. The HACO algorithm reduced the total costs and comprehensive carbon emission costs by 0.41% and 1.78% respectively, because the two improved strategies reinforced the exploration of optimal schemes and solutions. In addition, the computation time of the HACO algorithm was significantly shorter than that of the ACO algorithm. The reason might be interpreted as follows. The process of pheromone evaporation and deposition was applied universally to all paths in the ACO algorithm, thus increasing execution time and hindering the exploration of the solution space, and diminishing the convergence rate. In contrast, the HACO algorithm selectively reinforced pheromones on optimal and sub-optimal paths after evaporation. This targeted update reduced the influence of non-optimal paths and enhanced positive feedback, focused on search resources, and accelerated the convergence to the optimal solution. Consequently, HACO exhibited a significantly enhanced convergence rate and solution quality.

Table 8. Comparison of the results of ACO and HACO algorithms.

Types	Comprehensive Carbon Emission Costs	Total Costs	Time/s
ACO	1214.18	10,756.62	41.2
HACO	1192.57	10,712.48	0.04
Reduction rate	1.78%	0.41%	99.9%

5. Discussion and Conclusions

In the context of the rising prevalence of fresh food e-commerce, there is a pressing need for strategies to promote the sustainability of logistics operations while meeting the service demands of fresh food products. Therefore, this study focused on the joint optimization of the location-routing problem in a two-echelon cold chain network for the purposes to improve logistics efficiency, cost reduction, and sustainable logistics development. Based on the characteristics of perishable cargo and the requirements of low-carbon green logistics, the 2E-LCLRP model, considering both direct and indirect carbon emission costs, operational cost, fixed cost, transportation cost, refrigeration cost, and cargo damage cost, was proposed. A HACO algorithm was developed for the NP-hard problem, utilizing an adaptive probabilistic decision strategy and pheromone update rules. This approach was then validated in a practical application in Nanjing.

The main conclusions drawn are as follows:

1. In this work, HACO successfully overcame the disadvantage that ACO would fall into the local optimal solution. During the mid-stage of the algorithm's operation, ACO experienced stagnation for over 100 iterations. HACO used an adaptive hybrid selection strategy that effectively combined randomness with determinism, enhancing

its search capabilities. This method enabled the algorithm to swiftly jump out of the local optimal and avoid stagnation;

2. In terms of convergence speed, HACO demonstrated a significant advantage, requiring only 20 iterations to reach the optimal solution of the objective function. Furthermore, the comparative analysis indicated that HACO achieved up to a 99.9% reduction in execution time. This advantage was primarily attributable to the local pheromone updating strategy of HACO, which mitigated the impact of non-optimal solutions and rendered the algorithm more effective than the global pheromone updating strategy in terms of resource allocation and convergence speed;
3. HACO outperformed in finding high-quality solutions by adapting its search strategy based on solution characteristics throughout the iteration process and applying the pheromone updating strategy. The balance between exploration and exploitation led to optimal outcomes, with a focus on discovering the best path planning solution;
4. The results of the case study revealed that HACO, when compared with ACO, reduced total logistics cost, carbon emission cost, and execution time by 0.41%, 1.78%, and 99.9%, respectively. This indicated that HACO significantly outperformed the original ACO in balancing the environmental benefits and the logistics cost, thereby validated the efficacy of both the algorithm and the model.

HACO's flexibility and adaptability make it applicable to various logistics networks, including normal temperature logistics, pharmaceutical logistics, and multi-echelon logistics, as the networks encounter similar challenges in location-routing optimization. Consequently, this study could enhance the decision-making recommendations for enterprises in balancing economic and environmental costs. The result is conducive to the advancement of sustainable green logistics, facilitating enterprises in fulfilling their social responsibility and enhancing their brand image. The proposed model offers a novel research perspective for designing sustainable logistics networks, and the enhancement of HACO holds significant theoretical value for the resolution of complex optimization problems and the improvement of heuristic algorithms.

There are some limitations in this work. The proposed model does not consider the uncertainty for customer demand and instead relies on a number of simplistic assumptions, including fixed vehicle speeds. In view of this, future research should incorporate dynamic demand, seasonal factors, congestion indices, and real-time traffic data to improve the suitability for addressing complex optimization challenges. On the other hand, enhancing the algorithm's adaptability and stability is crucial for addressing more complex optimization challenges. Therefore, it is recommended to consider the optimization of heuristic algorithms in future work.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/math12121851/s1.

Author Contributions: Conceptualization, X.Z. and D.Z.; methodology, X.Z.; software, X.Z.; validation, X.Z.; formal analysis, X.Z.; investigation, X.Z. and Y.W.; resources, X.Z.; data curation, X.Z.; writing—original draft preparation, X.Z.; writing—review and editing, X.Z. and D.Z.; visualization, X.Z. and Y.W.; supervision, D.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by National College Student Innovation and Entrepreneurship Training Program grant number 202310307001Z.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in the study are included in the article and the Supplementary Material, further inquiries can be directed to the corresponding authors.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

2E-LCLRP	Two-echelon low-carbon location-routing problem
LRP	Location-routing problem
HACO	Hybrid ant colony optimization
IACO	Improved ant colony optimization
ACO	Ant colony optimization
EVs	Electric vehicles
UCW	Urban central warehouse
FW	Front warehouse

References

1. Qu, P.; Zhang, M.; Fan, K.; Guo, Z. Microporous modified atmosphere packaging to extend shelf life of fresh foods: A review. *Crit. Rev. Food Sci. Nutr.* **2022**, *62*, 51–65. [CrossRef] [PubMed]
2. Lin, J.; Li, T.; Guo, J. Factors influencing consumers' continuous purchase intention on fresh food e-commerce platforms: An organic foods-centric empirical investigation. *Electron. Commer. Res. Appl.* **2021**, *50*, 101103. [CrossRef]
3. Qin, G.; Tao, F.; Li, L. A Vehicle Routing Optimization Problem for Cold Chain Logistics Considering Customer Satisfaction and Carbon Emissions. *Int. J. Environ. Res. Public Health* **2019**, *16*, 576. [CrossRef]
4. Han, J.W.; Zuo, M.; Zhu, W.Y.; Zuo, J.H.; Lü, E.L.; Yang, X.T. A comprehensive review of cold chain logistics for fresh agricultural products: Current status, challenges, and future trends. *Trends Food Sci. Technol.* **2021**, *109*, 536–551. [CrossRef]
5. Liu, Z.; Huang, Y.Q.; Shang, W.L.; Zhao, Y.J.; Yang, Z.L.; Zhao, Z. Precooling energy and carbon emission reduction technology investment model in a fresh food cold chain based on a differential game. *Appl. Energy* **2022**, *326*, 119945. [CrossRef]
6. Abbasi, S.; Moosivand, M.; Vlachos, I.; Talooni, M. Designing the Location–Routing Problem for a Cold Supply Chain Considering the COVID-19 Disaster. *Sustainability* **2023**, *15*, 15490. [CrossRef]
7. Li, K.; Li, D.; Wu, D. Multi-objective Optimization for Location-Routing-Inventory Problem in Cold Chain Logistics Network with Soft Time Window Constraint. *J. Eur. Des Systèmes Autom.* **2020**, *53*, 803–809. [CrossRef]
8. Wu, X. The Front Warehouse Model of the "Missfresh" E-commerce Company Under the Background of Artificial Intelligence. In Proceedings of the 3rd International Conference on Artificial Intelligence in Electronics Engineering, Bangkok, Thailand, 11–13 January 2022; pp. 49–54. [CrossRef]
9. Li, Y.; Yang, J. The last-mile delivery vehicle routing problem with handling cost in the front warehouse mode. *Comput. Ind. Eng.* **2024**, *190*, 110076. [CrossRef]
10. Wang, Y.; Sun, Y.; Guan, X.; Guo, Y. Two-Echelon Location-Routing Problem with Time Windows and Transportation Resource Sharing. *J. Adv. Transp.* **2021**, *2021*, 8280686. [CrossRef]
11. Ziaei, Z.; Jabbarzadeh, A. A multi-objective robust optimization approach for green location-routing planning of multi-modal transportation systems under uncertainty. *J. Clean. Prod.* **2021**, *291*, 125293. [CrossRef]
12. Saif-Eddine, A.S.; El-Beheiry, M.M.; El-Kharbotly, A.K. An improved genetic algorithm for optimizing total supply chain cost in inventory location routing problem. *Ain Shams Eng. J.* **2019**, *10*, 63–76. [CrossRef]
13. Watson-Gandy, C.; Dohrn, P. Depot location with van salesmen—A practical approach. *Omega* **1973**, *1*, 321–329. [CrossRef]
14. Gao, S.; Wang, Y.; Cheng, J.; Inazumi, Y.; Tang, Z. Ant colony optimization with clustering for solving the dynamic location routing problem. *Appl. Math. Comput.* **2016**, *285*, 149–173. [CrossRef]
15. Xu, X.; Liu, W.; Jiang, M.; Lin, Z. A multi-cycle and multi-echelon location-routing problem for integrated reverse logistics. *Ind. Manag. Data Syst.* **2022**, *122*, 2237–2260. [CrossRef]
16. Aghighi, A.; Goli, A.; Malmir, B.; Tirkolaee, E.B. The stochastic location-routing-inventory problem of perishable products with reneging and balking. *J. Ambient Intell. Humaniz. Comput.* **2023**, *14*, 6497–6516. [CrossRef]
17. Tasouji Hassanpour, S.; Ke, G.Y.; Zhao, J.; Tulett, D.M. Infectious waste management during a pandemic: A stochastic location-routing problem with chance-constrained time windows. *Comput. Ind. Eng.* **2023**, *177*, 109066. [CrossRef]
18. Jiaojiao, W.; Yueyu, L.; Congcong, C. Study on Location and Route Optimization of Distribution Center of a Digital Company Based on Clustering Algorithm and Ant Colony Algorithm Method. In Proceedings of the 2020 9th International Conference on Industrial Technology and Management (ICITM), Oxford, UK, 11–13 February 2020; pp. 202–205. [CrossRef]
19. Leng, L.; Zhang, C.; Zhao, Y.; Wang, W.; Zhang, J.; Li, G. Biobjective low-carbon location-routing problem for cold chain logistics: Formulation and heuristic approaches. *J. Clean. Prod.* **2020**, *273*, 122801. [CrossRef]
20. Sun, J.; Tang, J.; Fu, W.; Chen, Z.; Niu, Y. Construction of a multi-echelon supply chain complex network evolution model and robustness analysis of cascading failure. *Comput. Ind. Eng.* **2020**, *144*, 106457. [CrossRef]
21. Dai, Z.; Aqlan, F.; Gao, K.; Zhou, Y. A two-phase method for multi-echelon location-routing problems in supply chains. *Expert Syst. Appl.* **2019**, *115*, 618–634. [CrossRef]
22. Mirhedayatian, S.M.; Crainic, T.G.; Guajardo, M.; Wallace, S.W. A two-echelon location-routing problem with synchronisation. *J. Oper. Res. Soc.* **2021**, *72*, 145–160. [CrossRef]

23. Fazayeli, S.; Eydi, A.; Kamalabadi, I.N. A model for distribution centers location-routing problem on a multimodal transportation network with a meta-heuristic solving approach. *J. Ind. Eng. Int.* **2018**, *14*, 327–342. [CrossRef]
24. Darvish, M.; Archetti, C.; Coelho, L.C.; Speranza, M.G. Flexible two-echelon location routing problem. *Eur. J. Oper. Res.* **2019**, *277*, 1124–1136. [CrossRef]
25. Chen, J.; Liao, W.; Yu, C. Route optimization for cold chain logistics of front warehouses based on traffic congestion and carbon emission. *Comput. Ind. Eng.* **2021**, *161*, 107663. [CrossRef]
26. Wu, X. Optimization of Distribution Mode of the Fresh Food Logistics Front Warehouse from the Perspective of Intelligent Logistics. In Proceedings of the 2021 IEEE Conference on Telecommunications, Optics and Computer Science (TOCS), Shenyang, China, 10–11 December 2021; pp. 307–311. [CrossRef]
27. Zhu, J.; Tian, X. Value of High-Quality Distribution in Front Warehouse Mode Retailing. *Procedia Comput. Sci.* **2022**, *199*, 110–117. [CrossRef]
28. Tang, J.; Qi, C.; Wang, H. Integrated optimization of order splitting and distribution routing for the front warehouse mode e-retailing. *Int. J. Prod. Res.* **2023**, *62* , 1715–1736. [CrossRef]
29. Chen, W.; Zhang, Y.; Zhou, Y. Integrated scheduling of zone picking and vehicle routing problem with time windows in the front warehouse mode. *Comput. Ind. Eng.* **2022**, *163*, 107823. [CrossRef]
30. Wang, Y.; Peng, S.; Zhou, X.; Mahmoudi, M.; Zhen, L. Green logistics location-routing problem with eco-packages. *Transp. Res. Part E Logist. Transp. Rev.* **2020**, *143*, 102118. [CrossRef]
31. Yingfei, Y.; Mengze, Z.; Zeyu, L.; Ki-Hyung, B.; Andriandafiarisoa Ralison Ny Avotra, A.; Nawaz, A. Green logistics performance and infrastructure on service trade and environment-Measuring firm's performance and service quality. *J. King Saud Univ. Sci.* **2022**, *34*, 101683. [CrossRef]
32. Dzwigo, H.; Trushkina, N.; Kwilinski, A. The Organizational and Economic Mechanism of Implementing the Concept of Green Logistics. *Virtual Econ.* **2021**, *4*, 41–75. [CrossRef]
33. Yao, X.; Cheng, Y.; Song, M. Assessment of collaboration in city logistics: From the aspects of profit and CO_2 emissions. *Int. J. Logist. Res. Appl.* **2019**, *22*, 576–591. [CrossRef]
34. Zhang, S.; Chen, N.; She, N.; Li, K. Location optimization of a competitive distribution central for urban cold chain logistics in terms of low-carbon emissions. *Comput. Ind. Eng.* **2021**, *154*, 107120. [CrossRef]
35. Wang, Z.; Leng, L.; Wang, S.; Li, G.; Zhao, Y. A Hyperheuristic Approach for Location-Routing Problem of Cold Chain Logistics considering Fuel Consumption. *Comput. Intell. Neurosci.* **2020**, *2020*, 8395754. [CrossRef] [PubMed]
36. Dukkanci, O.; Kara, B.Y.; Bektaş, T. The green location-routing problem. *Comput. Oper. Res.* **2019**, *105*, 187–202. [CrossRef]
37. Zhang, C.; Zhao, Y.; Leng, L. A Hyper-Heuristic Algorithm for Time-Dependent Green Location Routing Problem With Time Windows. *IEEE Access* **2020**, *8*, 83092–83104. [CrossRef]
38. Li, Y.; Lim, M.K.; Hu, J.; Tseng, M.L. Investigating the effect of carbon tax and carbon quota policy to achieve low carbon logistics operations. *Resour. Conserv. Recycl.* **2020**, *154*, 104535. [CrossRef]
39. Wang, S.; Tao, F.; Shi, Y. Optimization of Location–Routing Problem for Cold Chain Logistics Considering Carbon Footprint. *Int. J. Environ. Res. Public Health* **2018**, *15*, 86. [CrossRef] [PubMed]
40. Taptich, M.N.; Scown, C.D.; Piscopo, K.; Horvath, A. Drop-in biofuels offer strategies for meeting California's 2030 climate mandate. *Environ. Res. Lett.* **2018**, *13*, 094018. [CrossRef]
41. Grahn, M.; Azar, C.; Williander, M.I.; Anderson, J.E.; Mueller, S.A.; Wallington, T.J. Fuel and Vehicle Technology Choices for Passenger Vehicles in Achieving Stringent CO_2 Targets: Connections between Transportation and Other Energy Sectors. *Environ. Sci. Technol.* **2009**, *43*, 3365–3371. [CrossRef] [PubMed]
42. Bonsu, N.O. Towards a circular and low-carbon economy: Insights from the transitioning to electric vehicles and net zero economy. *J. Clean. Prod.* **2020**, *256*, 120659. [CrossRef]
43. Dall-Orsoletta, A.; Ferreira, P.; Gilson Dranka, G. Low-carbon technologies and just energy transition: Prospects for electric vehicles. *Energy Convers. Manag. X* **2022**, *16*, 100271. [CrossRef]
44. Nilsson, M.; Nykvist, B. Governing the electric vehicle transition – Near term interventions to support a green energy economy. *Appl. Energy* **2016**, *179*, 1360–1371. [CrossRef]
45. Wu, T.; Li, Z.; Wang, G.; Zhang, X.; Qiu, J. Low-Carbon Charging Facilities Planning for Electric Vehicles Based on a Novel Travel Route Choice Model. *IEEE Trans. Intell. Transp. Syst.* **2023**, *24*, 5908–5922. [CrossRef]
46. Almouhanna, A.; Quintero-Araujo, C.L.; Panadero, J.; Juan, A.A.; Khosravi, B.; Ouelhadj, D. The location routing problem using electric vehicles with constrained distance. *Comput. Oper. Res.* **2020**, *115*, 104864. [CrossRef]
47. Çatay, B.; Sadati, İ. An improved matheuristic for solving the electric vehicle routing problem with time windows and synchronized mobile charging/battery swapping. *Comput. Oper. Res.* **2023**, *159*, 106310. [CrossRef]
48. Lee, C. An exact algorithm for the electric-vehicle routing problem with nonlinear charging time. *J. Oper. Res. Soc.* **2021**, *72*, 1461–1485. [CrossRef]
49. Chen, Y.; Li, D.; Zhang, Z.; Wahab, M.; Jiang, Y. Solving the battery swap station location-routing problem with a mixed fleet of electric and conventional vehicles using a heuristic branch-and-price algorithm with an adaptive selection scheme. *Expert Syst. Appl.* **2021**, *186*, 115683. [CrossRef]
50. Wang, M.; Miao, L.; Zhang, C. A branch-and-price algorithm for a green location routing problem with multi-type charging infrastructure. *Transp. Res. Part E Logist. Transp. Rev.* **2021**, *156*, 102529. [CrossRef]

51. Arslan, O.; Karaşan, O.E.; Mahjoub, A.R.; Yaman, H. A Branch-and-Cut Algorithm for the Alternative Fuel Refueling Station Location Problem with Routing. *Transp. Sci.* **2019**, *53*, 1107–1125. [CrossRef]
52. Bao, Z.; Xie, C. Optimal station locations for en-route charging of electric vehicles in congested intercity networks: A new problem formulation and exact and approximate partitioning algorithms. *Transp. Res. Part C Emerg. Technol.* **2021**, *133*, 103447. [CrossRef]
53. Christensen, T.R.L.; Klose, A. A fast exact method for the capacitated facility location problem with differentiable convex production costs. *Eur. J. Oper. Res.* **2021**, *292*, 855–868. [CrossRef]
54. Rafie-Majd, Z.; Pasandideh, S.H.R.; Naderi, B. Modelling and solving the integrated inventory-location-routing problem in a multi-period and multi-perishable product supply chain with uncertainty: Lagrangian relaxation algorithm. *Comput. Chem. Eng.* **2018**, *109*, 9–22. [CrossRef]
55. Biuki, M.; Kazemi, A.; Alinezhad, A. An integrated location-routing-inventory model for sustainable design of a perishable products supply chain network. *J. Clean. Prod.* **2020**, *260*, 120842. [CrossRef]
56. De Oliveira, S.M.; Bezerra, L.C.; Stützle, T.; Dorigo, M.; Wanner, E.F.; De Souza, S.R. A computational study on ant colony optimization for the traveling salesman problem with dynamic demands. *Comput. Oper. Res.* **2021**, *135*, 105359. [CrossRef]
57. Dewantoro, R.W.; Sihombing, P. The Combination of Ant Colony Optimization (ACO) and Tabu Search (TS) Algorithm to Solve the Traveling Salesman Problem (TSP). In Proceedings of the 2019 3rd International Conference on Electrical, Telecommunication and Computer Engineering (ELTICOM), Medan, Indonesia, 16–17 September 2019; pp. 160–164. [CrossRef]
58. Tuani, A.F.; Keedwell, E.; Collett, M. Heterogenous Adaptive Ant Colony Optimization with 3-opt local search for the Travelling Salesman Problem. *Appl. Soft Comput.* **2020**, *97*, 106720. [CrossRef]
59. Jia, Y.H.; Mei, Y.; Zhang, M. A Bilevel Ant Colony Optimization Algorithm for Capacitated Electric Vehicle Routing Problem. *IEEE Trans. Cybern.* **2022**, *52*, 10855–10868. [CrossRef]
60. Xiang, X.; Qiu, J.; Xiao, J.; Zhang, X. Demand coverage diversity based ant colony optimization for dynamic vehicle routing problems. *Eng. Appl. Artif. Intell.* **2020**, *91*, 103582. [CrossRef]
61. Cao, R.; Li, S.; Ji, Y.; Zhang, Z.; Xu, H.; Zhang, M.; Li, M.; Li, H. Task assignment of multiple agricultural machinery cooperation based on improved ant colony algorithm. *Comput. Electron. Agric.* **2021**, *182*, 105993. [CrossRef]
62. Deng, W.; Xu, J.; Zhao, H. An Improved Ant Colony Optimization Algorithm Based on Hybrid Strategies for Scheduling Problem. *IEEE Access* **2019**, *7*, 20281–20292. [CrossRef]
63. Guan, B.; Zhao, Y.; Li, Y. An improved ant colony optimization with an automatic updating mechanism for constraint satisfaction problems. *Expert Syst. Appl.* **2021**, *164*, 114021. [CrossRef]
64. Wang, M.; Ma, T.; Li, G.; Zhai, X.; Qiao, S. Ant Colony Optimization With an Improved Pheromone Model for Solving MTSP With Capacity and Time Window Constraint. *IEEE Access* **2020**, *8*, 106872–106879. [CrossRef]
65. Jiang, J.; Yu, N.; Ye, J.; Bai, W. Vehicle logistics path optimization based on ant colony and particle hybrid algorithm. *J. Phys. Conf. Ser.* **2021**, *1865*, 042086. [CrossRef]
66. Shi, K.; Huang, L.; Jiang, D.; Sun, Y.; Tong, X.; Xie, Y.; Fang, Z. Path Planning Optimization of Intelligent Vehicle Based on Improved Genetic and Ant Colony Hybrid Algorithm. *Front. Bioeng. Biotechnol.* **2022**, *10*, 905983. [CrossRef]
67. Wang, Y.; Wang, L.; Chen, G.; Cai, Z.; Zhou, Y.; Xing, L. An Improved Ant Colony Optimization algorithm to the Periodic Vehicle Routing Problem with Time Window and Service Choice. *Swarm Evol. Comput.* **2020**, *55*, 100675. [CrossRef]
68. Di Caprio, D.; Ebrahimnejad, A.; Alrezaamiri, H.; Santos-Arteaga, F.J. A novel ant colony algorithm for solving shortest path problems with fuzzy arc weights. *Alex. Eng. J.* **2022**, *61*, 3403–3415. [CrossRef]
69. Goel, R.; Maini, R. A hybrid of ant colony and firefly algorithms (HAFA) for solving vehicle routing problems. *J. Comput. Sci.* **2018**, *25*, 28–37. [CrossRef]
70. Hou, W.; Xiong, Z.; Wang, C.; Chen, H. Enhanced ant colony algorithm with communication mechanism for mobile robot path planning. *Robot. Auton. Syst.* **2022**, *148*, 103949. [CrossRef]
71. Li, Y.; Soleimani, H.; Zohal, M. An improved ant colony optimization algorithm for the multi-depot green vehicle routing problem with multiple objectives. *J. Clean. Prod.* **2019**, *227*, 1161–1172. [CrossRef]
72. He, M.; Wei, Z.; Wu, X.; Peng, Y. An Adaptive Variable Neighborhood Search Ant Colony Algorithm for Vehicle Routing Problem With Soft Time Windows. *IEEE Access* **2021**, *9*, 21258–21266. [CrossRef]
73. Liang, S.; Jiao, T.; Du, W.; Qu, S. An improved ant colony optimization algorithm based on context for tourism route planning. *PLoS ONE* **2021**, *16*, e0257317. [CrossRef]
74. Guo, K.; Hu, S.; Zhu, H.; Tan, W. Industrial information integration method to vehicle routing optimization using grey target decision. *J. Ind. Inf. Integr.* **2022**, *27*, 100336. [CrossRef]
75. Wang, X.; Zhan, L.; Zhang, Y.; Fei, T.; Tseng, M.L. Environmental cold chain distribution central location model in the semiconductor supply chain: A hybrid arithmetic whale optimization algorithm. *Comput. Ind. Eng.* **2024**, *187*, 109773. [CrossRef]
76. Horvat, M.; Jović, A.; Burnik, K. Assessing the Robustness of Cluster Solutions in Emotionally-Annotated Pictures Using Monte-Carlo Simulation Stabilized K-Means Algorithm. *Mach. Learn. Knowl. Extr.* **2021**, *3*, 435–452. [CrossRef]
77. Zheng, F.; Meng, X.; Wang, L.; Zhang, N. Operation Optimization Method of Distribution Network with Wind Turbine and Photovoltaic Considering Clustering and Energy Storage. *Sustainability* **2023**, *15*, 2184. [CrossRef]
78. Zhao, B.; Gui, H.; Li, H.; Xue, J. Cold Chain Logistics Path Optimization via Improved Multi-Objective Ant Colony Algorithm. *IEEE Access* **2020**, *8*, 142977–142995. [CrossRef]

79. Chen, A.; Tan, H.; Zhu, Y. Ant colony optimization algorithm and its application. In Proceedings of the 2nd International Conference on Applied Mathematics, Modelling, and Intelligent Computing (CAMMIC 2022), Kunming, China, 25–27 March 2022; p. 265. [CrossRef]
80. Ebadinezhad, S. DEACO: Adopting dynamic evaporation strategy to enhance ACO algorithm for the traveling salesman problem. *Eng. Appl. Artif. Intell.* **2020**, *92*, 103649. [CrossRef]
81. Wu, H.; Gao, Y.; Wang, W.; Zhang, Z. A hybrid ant colony algorithm based on multiple strategies for the vehicle routing problem with time windows. *Complex Intell. Syst.* **2023**, *9*, 2491–2508. [CrossRef]
82. Chen, J.; Gui, P.; Ding, T.; Na, S.; Zhou, Y. Optimization of Transportation Routing Problem for Fresh Food by Improved Ant Colony Algorithm Based on Tabu Search. *Sustainability* **2019**, *11*, 6584. [CrossRef]
83. Wu, L.; Huang, X.; Cui, J.; Liu, C.; Xiao, W. Modified adaptive ant colony optimization algorithm and its application for solving path planning of mobile robot. *Expert Syst. Appl.* **2023**, *215*, 119410. [CrossRef]
84. Liu, J.; Liu, J. Applying multi-objective ant colony optimization algorithm for solving the unequal area facility layout problems. *Appl. Soft Comput.* **2019**, *74*, 167–189. [CrossRef]
85. Yi, N.; Xu, J.; Yan, L.; Huang, L. Task optimization and scheduling of distributed cyber–physical system based on improved ant colony algorithm. *Future Gener. Comput. Syst.* **2020**, *109*, 134–148. [CrossRef]
86. Agramanisti Azdy, R.; Darnis, F. Use of Haversine Formula in Finding Distance Between Temporary Shelter and Waste End Processing Sites. *J. Phys. Conf. Ser.* **2020**, *1500*, 012104. [CrossRef]
87. Dauni, P.; Firdaus, M.D.; Asfariani, R.; Saputra, M.I.N.; Hidayat, A.A.; Zulfikar, W.B. Implementation of Haversine formula for school location tracking. *J. Phys. Conf. Ser.* **2019**, *1402*, 077028. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.